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Roll No. .....

TEC–403

B. TECH. (EC) (FOURTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

MICROCONTROLLER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write the features of microprocessor 8085. Discuss address bus, data bus, and control bus. (CO1)

OR

(b) Interface a 2 KB RAM with 8085 keeping the starting address as 4000H. Also find its last address. (CO1)

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2. (a) What is the difference between programmed I/O and interrupt driven I/O ? Explain the DMA technique. (CO1)

OR

(b) Give the format of flag register of 8085. Determine the status of all 8085 flags when it adds 85H and 7AH. (CO1)

3. (a) Draw the pin diagram of 8051. Explain the function of ALE, RD, WR, RST pins of 8051. (CO2)

OR

(b) Give answers of following questions pertaining to the internal RAM of 8051. (CO2)

(i) What is the total size and address range of internal RAM ?

(ii) What is the address range of bit addressable space ?

(iii) Name the registers present in Bank 1.

(iv) What is the default value of stack pointer ?

(v) Give the address range of registers of Bank 2.

4. (a) Differentiate between : (CO2)

(i) Harvard and Von-Neumann architecture

(ii) CISC and RISC

OR

(b) Discuss the function of the following registers of 8051 : (CO2)

(i) PC

(ii) SP

(iii) A

(iv) B

(v) DPTR

5. (a) Discuss 8051 addressing modes with examples. (CO3)

OR

(b) Write an 8051 program to move a block of 10 bytes from starting address 6050H to 6060H. (CO3)